

Experiment 06

VLAN (Virtual Local Area Network) and VLAN Trunking

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Aim

To study the concept of Virtual LANs (VLANs) and VLAN Trunking, to understand how a single physical switch can be logically divided into multiple independent broadcast domains, and to configure and verify VLAN segmentation together with IEEE 802.1Q trunking between two switches using Cisco Packet Tracer.

Objectives

- Understand how a Layer 2 switch forwards frames using its MAC address table.
- Differentiate between a Layer 2 switch and a Layer 3 switch.
- Understand the concept of a VLAN and why it is used to separate broadcast domains.
- Differentiate between an access port and a trunk port.
- Understand VLAN tagging using the IEEE 802.1Q standard.
- Configure VLANs on a single switch and verify inter-VLAN isolation.
- Configure a trunk link between two switches and verify that a single VLAN can span across them.

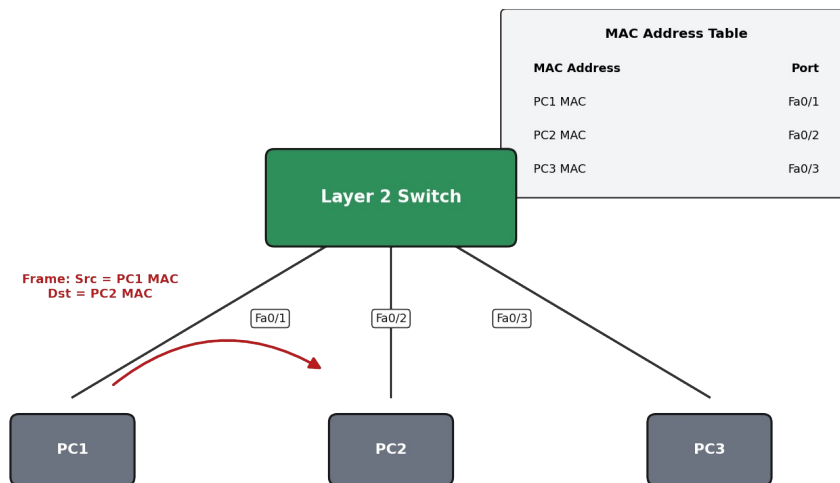
Software / Tools Required

- Cisco Packet Tracer
- Windows Operating System

Theory

A) Layer 2 Switch and the MAC Address Table

A Layer 2 switch is a networking device that operates at the Data Link Layer of the OSI model and forwards Ethernet frames on the basis of MAC addresses. When PC1 wants to talk to PC2, it sends a frame containing PC1's MAC address as the source and PC2's MAC address as the destination. The switch inspects the destination MAC address, looks it up in a table it maintains internally — the MAC Address Table — and forwards the frame only out of the port on which that address was learnt. This lookup-and-forward behaviour is the fundamental operation of every Layer 2 switch, including the Cisco 2960 series used throughout this experiment.



The switch inspects the destination MAC address and forwards the frame only out of Fa0/2 (PC2's port).

Fig. 6.1 — A Layer 2 switch forwards a frame only to the port associated with the destination MAC address

B) Layer 2 Switch vs Layer 3 Switch

A Layer 3 switch combines ordinary Layer 2 switching with Layer 3 (IP-based) routing capability. Where a plain Layer 2 switch can only forward frames using MAC addresses, a Layer 3 switch can also route packets between different IP subnets — which is what makes Inter-VLAN routing possible without a separate external router.

Feature	Layer 2 Switch	Layer 3 Switch
Primary OSI Layer	Layer 2	Layer 2 + Layer 3
Uses MAC address	Yes	Yes
Uses IP address for routing	No	Yes
Switching within a VLAN	Yes	Yes
Routing between VLANs	No	Yes (Inter-VLAN Routing)
Typical example	Cisco 2960	Cisco 3560

C) What is a VLAN?

VLAN stands for Virtual Local Area Network. Without VLANs, every device connected to a physical switch belongs to the same single broadcast domain. A VLAN allows a single physical switch to be logically split into multiple independent broadcast domains — for example, one for the HR department and another for the Sales department — without needing separate physical switches for each department. Devices in different VLANs cannot communicate with one another unless a router or a Layer 3 switch performs Inter-VLAN routing between them.

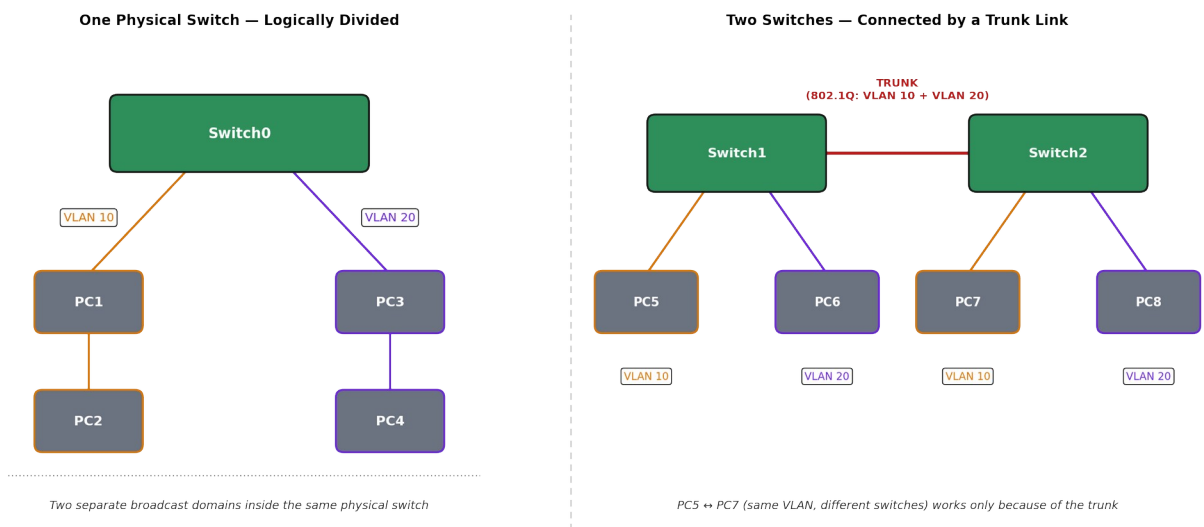


Fig. 6.2 — Left: one physical switch logically divided into VLAN 10 and VLAN 20. Right: two switches, each with members of both VLANs, joined by a trunk link

D) Why Do VLANs Separate Devices?

A switch normally floods broadcast traffic to every port within a broadcast domain. Once VLANs are created, the switch logically separates its ports into different broadcast domains, so a broadcast generated inside VLAN 10 is not delivered to any port belonging to VLAN 20, and vice-versa — even though both VLANs may be running on the very same physical switch.

E) Limitations of a Traditional (Non-VLAN) LAN and How VLANs Address Them

Limitation of a Flat LAN	How VLANs Help
Every device shares one large broadcast domain.	A physical LAN is split into several smaller logical broadcast domains, cutting down unnecessary broadcast traffic.
Departments are hard to keep separate.	Each department (e.g. Accounts, HR, IT) can be placed in its own VLAN.
Weak isolation between users.	Devices in different VLANs cannot exchange traffic directly unless routing is explicitly configured.
Broadcasts reach devices that do not need them.	Broadcasts stay confined to their own VLAN, improving overall network performance.
Network grouping is tied to physical location.	VLAN membership is logical, so two PCs in different rooms (or floors) can still belong to the same VLAN.
Relocating a user often means re-cabling.	Moving a user to a new group is just a matter of re-assigning their switch port to a different VLAN.
A large LAN becomes hard to manage.	Splitting it into smaller VLANs keeps the network organised and easier to troubleshoot.
Extra switches are needed for every extra department.	Multiple logical networks can share the same physical switch hardware, saving cost.

F) Access Ports and Trunk Ports

An access port is a switch port that belongs to exactly one VLAN and is normally used to connect end devices such as PCs, printers or servers — traffic on an access port is untagged. When two switches both host the same VLANs, a single access-port-style link cannot carry more than one VLAN, so a special port called a trunk port is used

instead. A trunk port can carry traffic belonging to multiple VLANs over a single physical link, which is exactly what connects Switch1 and Switch2 in Part B of this experiment.

Access Port	Trunk Port
Normally carries a single VLAN.	Carries multiple VLANs over one link.
Usually connects an end device.	Usually connects two network devices (switch-to-switch or switch-to-router).
Example: PC – Switch.	Example: Switch – Switch.
Traffic is untagged toward the end device. e.g. Fa0/1 assigned to VLAN 10.	VLAN membership is identified using a VLAN tag. e.g. Gi0/1 carrying VLAN 10 and VLAN 20 together.

G) VLAN Tagging — IEEE 802.1Q

When a frame crosses a trunk link, the receiving switch has no way of knowing which VLAN the frame belongs to unless that information travels with the frame. This is solved by VLAN tagging: as a frame leaves over a trunk, the sending switch inserts a small tag into the Ethernet frame that records the VLAN ID (for example, VLAN ID = 10). The receiving switch reads this tag, learns that the frame belongs to VLAN 10, strips the tag if the destination is an access port, and forwards it only to ports that belong to VLAN 10. The industry-standard tagging mechanism used for this is IEEE 802.1Q.

H) VLAN Trunking vs 802.1Q vs VTP — Avoiding Confusion

Three related terms are often confused with one another. VLAN Trunking is the general concept of carrying multiple VLANs across a single physical link. IEEE 802.1Q is the actual open standard used to tag frames so that VLAN information survives the trip across a trunk. VTP (VLAN Trunking Protocol), despite its name, does something different — it is used to distribute and synchronise VLAN configuration information across multiple Cisco switches within the same VTP domain, and is not itself responsible for carrying multi-VLAN traffic over a link.

Trunking Standard	Type	Status / Use
IEEE 802.1Q	Open industry standard	Current, most widely used VLAN trunking / tagging method
ISL (Inter-Switch Link)	Cisco proprietary	Legacy, largely obsolete on modern equipment

IEEE 802.1Q trunking was used throughout this experiment.

Procedure

Part A: VLAN Segmentation on a Single Switch

A single switch was used to host two departments — HR and Sales — as two separate VLANs, to demonstrate that devices in different VLANs cannot reach each other even though they share the same physical switch.

Device	VLAN	Switch Port	IP Address
PC0	10 (hr)	Fa0/1	192.168.1.1
PC1	10 (hr)	Fa0/2	192.168.1.2
PC2	20 (sale)	Fa0/3	192.168.2.1
PC3	20 (sale)	Fa0/4	192.168.2.2

1. Place one 2960-24TT switch (Switch0) and four PCs in the Packet Tracer workspace, and connect PC0–PC3 to the switch using copper straight-through cables as per the table above.
2. Assign the IP addresses shown above to PC0, PC1 (HR / 192.168.1.0 /24) and PC2, PC3 (Sales / 192.168.2.0 /24).
3. Open Switch0's CLI and enter privileged and global configuration mode: enable, then configure terminal.
4. Create the two VLANs: vlan 10 followed by name hr, then vlan 20 followed by name sale.

- Assign the access ports to their VLANs: on interface fa0/1 and fa0/2, run switchport mode access and switchport access vlan 10; on interface fa0/3 and fa0/4, run switchport mode access and switchport access vlan 20.
- Verify the configuration using show vlan brief, and test connectivity: PC0 → PC1 and PC2 → PC3 should succeed (same VLAN), while PC0 → PC2 should fail (different VLANs, no routing configured).

Part B: VLAN Trunking Between Two Switches

Since a real organisation rarely fits on one switch, the same two VLANs — hr and sale — were then spread across two separate switches, one member of each department per switch, connected together by a single 802.1Q trunk link, to demonstrate that a trunk allows a VLAN to legitimately span multiple physical switches.

Switch	Device	VLAN	Access Port	IP Address
Switch1	PC4	10 (hr)	Fa0/1	192.168.1.1
Switch1	PC5	20 (sale)	Fa0/3	192.168.2.1
Switch2	PC7	10 (hr)	Fa0/1	192.168.1.2
Switch2	PC6	20 (sale)	Fa0/3	192.168.2.2

- Place two 2960-24TT switches (Switch1, Switch2) and four PCs. Connect PC4 and PC5 to Switch1, and PC7 and PC6 to Switch2, using straight-through cables; connect Switch1 Gi0/1 to Switch2 Gi0/1 to form the inter-switch link.
- Assign IP addresses to all four PCs as shown above — note that hr keeps the 192.168.1.0 /24 range and sale keeps 192.168.2.0 /24, regardless of which physical switch the PC is attached to.
- On Switch1, create the VLANs and assign the access ports, exactly as in Part A: vlan 10 / name hr, vlan 20 / name sale, then interface fa0/1 → access → vlan 10, and interface fa0/3 → access → vlan 20.
- Configure the inter-switch link as a trunk: interface gigabitEthernet0/1, then switchport mode trunk. This is the single most important step of the experiment — without it, the link would only be able to carry one VLAN's traffic, or none at all.
- Repeat the identical VLAN and trunk configuration on Switch2.
- Verify the VLAN table before and after the trunk is configured using show vlan, confirm the trunk is active using show interfaces trunk, and inspect learnt addresses using show mac address-table.
- Test connectivity: PC4 → PC7 (both hr, different switches) and PC5 → PC6 (both sale, different switches) should succeed because the trunk carries both VLANs; PC4 → PC5 (different VLANs) should still fail, since no Inter-VLAN routing has been configured.

Reference Configuration — Switch1

```
enable
configure terminal
vlan 10
  name hr
exit
vlan 20
  name sale
exit
interface fa0/1
  switchport mode access
  switchport access vlan 10
exit
interface fa0/3
  switchport mode access
  switchport access vlan 20
exit
```

```

interface gigabitEthernet0/1
  switchport mode trunk
exit
    
```

The same configuration (with fa0/1 carrying PC7 and fa0/3 carrying PC6) was repeated on Switch2.

Output

The completed topologies for Part A (single-switch VLAN segmentation) and Part B (two switches joined by a trunk) are shown below, as built and tested in Cisco Packet Tracer.

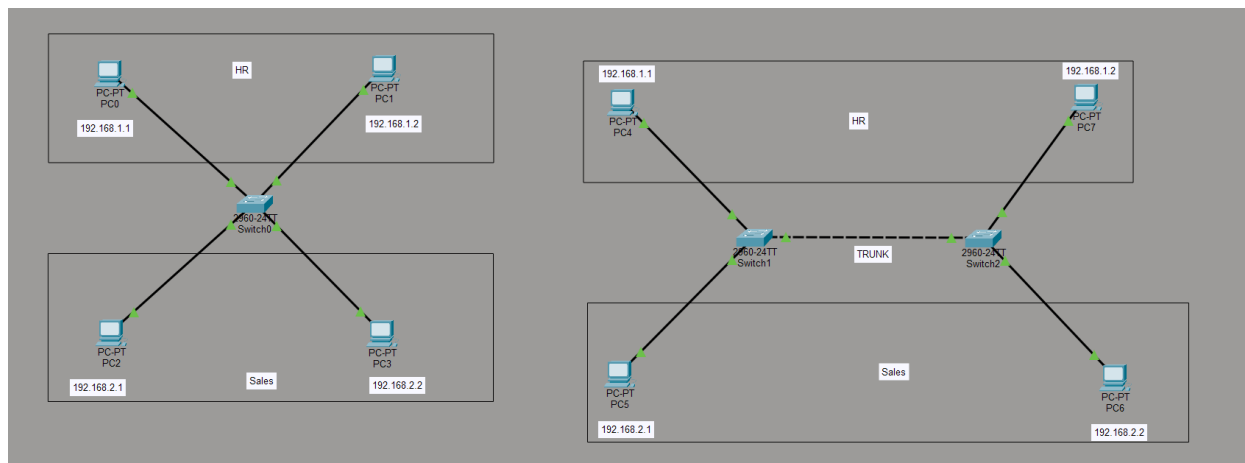


Fig. 6.3 — Left: HR and Sales as two VLANs on a single switch (Switch0). Right: HR and Sales spread across Switch1 and Switch2, joined by a Gi0/1–Gi0/1 trunk

The CLI output below, captured on one of the switches, shows the VLAN table before and after the trunk was configured on Gi0/1. Notice that VLAN 10 is named hr (port Fa0/1) and VLAN 20 is named sale (port Fa0/3), exactly as configured; and that once Gi0/1 is converted into a trunk, it is removed from VLAN 1's list of access ports, since a trunk port no longer belongs to a single VLAN.

```

VLAN Name                Status    Ports
-----
1    default                active    Fa0/2, Fa0/4, Fa0/5, Fa0/6
                                           Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                           Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                           Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                           Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                           Fa0/23, Fa0/24, Gig0/1, Gig0/2

10   hr                     active    Fa0/1
20   sale                   active    Fa0/3
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active

Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface gig0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
sh vlan

VLAN Name                Status    Ports
-----
1    default                active    Fa0/2, Fa0/4, Fa0/5, Fa0/6
                                           Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                           Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                           Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                           Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                           Fa0/23, Fa0/24, Gig0/2

10   hr                     active    Fa0/1
20   sale                   active    Fa0/3
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active

VLAN Type  SAID      MTU    Parent RingNo BridgeNo Stp    BrgdMode Trans1 Trans2
-----
1    enet    100001    1500    -      -      -      -      -      0      0
10   enet    100010    1500    -      -      -      -      -      0      0
20   enet    100020    1500    -      -      -      -      -      0      0
1002 fddi    101002    1500    -      -      -      -      -      0      0

Switch#
    
```

Fig. 6.4 — show vlan output before and after configuring Gi0/1 as an 802.1Q trunk (switchport mode trunk)

Observation

- In Part A, PC0 could reach PC1 and PC2 could reach PC3 (same VLAN), but PC0 could not reach PC2 (different VLANs on the same switch).
- In Part B, PC4 could reach PC7 and PC5 could reach PC6, even though each pair sits on a different physical switch, because the Gi0/1 trunk carried both VLAN 10 and VLAN 20 tagged with 802.1Q.
- PC4 could still not reach PC5, confirming that a trunk carries multiple VLANs side-by-side without merging them — devices in different VLANs remain isolated unless Inter-VLAN routing is added.
- Once Gi0/1 was converted to a trunk, it dropped out of VLAN 1's own port list in the show vlan output, confirming that a trunk port is no longer tied to a single access VLAN.

Result

VLAN segmentation and VLAN trunking were successfully studied and implemented using Cisco Packet Tracer. Two VLANs (hr and sale) were created and verified to be isolated from each other on a single switch, and were then successfully extended across two switches using an IEEE 802.1Q trunk link on Gi0/1, allowing devices in the same VLAN to communicate regardless of which physical switch they were connected to.

Learning Outcomes

- Understood how a Layer 2 switch forwards frames using its MAC address table, and how this differs from the routing behaviour of a Layer 3 switch.
- Understood the concept of a VLAN as a logical broadcast domain, independent of physical switch boundaries.
- Learned to create VLANs and assign access ports to them using the switchport access vlan command.

- Understood the difference between an access port and a trunk port, and why a trunk is required when a VLAN spans more than one switch.
- Learned how IEEE 802.1Q tagging allows a receiving switch to identify which VLAN a frame on a trunk belongs to.
- Understood the distinction between VLAN trunking (the concept), 802.1Q (the tagging standard), and VTP (a configuration-distribution protocol) — three related but different ideas.
- Verified VLAN and trunk configuration practically using show vlan, show interfaces trunk and show mac address-table, and confirmed inter-VLAN isolation and intra-VLAN connectivity using ping.